

FA3

Aldwin Benedict Montilla

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GitHub link: <https://github.com/aldwinmontilla661-dot/Aldwin-Benedict-Montilla/tree/main/FA3>

Question 5: Sensor Relevance Analysis

A geospatial analysis system has four sensors supplying images. The percentage of images supplied by each sensor and the percentage of images relevant to a query are shown in the following table.

Sensor	Percentage of Images Supplied	Percentage of Relevant Images
1	15	50
2	20	60
3	25	80
4	40	85

What is the overall percentage of relevant images?

R Code & Solution

```
# Input the data into data vectors
sensor <- c(1, 2, 3, 4)
pct_supplied <- c(15, 20, 25, 40)
pct_relevant <- c(50, 60, 80, 85)

# Create a clean data frame to hold the info
sensor_data <- data.frame(Sensor = sensor,
                          Pct_Supplied = pct_supplied,
                          Pct_Relevant = pct_relevant)

# Convert percentages to proportions for probability calculation
p_sensor <- sensor_data$Pct_Supplied / 100
p_relevant_given_sensor <- sensor_data$Pct_Relevant / 100

# Calculate the overall probability using the Law of Total Probability
# P(Relevant) = Sum of [P(Sensor_i) * P(Relevant | Sensor_i)]
overall_prob_relevant <- sum(p_sensor * p_relevant_given_sensor)
overall_prob_relevant

## [1] 0.735
```

```
# Convert back to a clean percentage format
overall_pct_relevant <- overall_prob_relevant * 100

# Print the result
print(paste0("The overall percentage of relevant images is: ", overall_pct_relevant, "%"))

## [1] "The overall percentage of relevant images is: 73.5%"
```

Interpretation

The total percentage of relevant images across the entire network of four sensors evaluates to **73.5 %**.

Question 6: Probability Independence Verification

A fair coin is tossed twice. Let the sample space be $S = HH, HT, TH, TT$, where each outcome is equally likely. Consider the following three events:

- E_1 : The outcomes of the two tosses match (HH or TT).
- E_2 : The first toss results in Heads (HH or HT).
- E_3 : The second toss results in Heads (TH or HH).

Show whether these three events are **pairwise independent** but not **mutually independent**.

```
# Define the complete sample space (S)
S <- c("HH", "HT", "TH", "TT")

# Define each event (E)
E1 <- c("HH", "TT")
E2 <- c("HH", "HT")
E3 <- c("TH", "HH")

# Create a custom function to compute probabilities
get_prob <- function(event_set) {
  length(event_set) / length(S)
}
```

```
# Compute individual probabilities
p_E1 <- get_prob(E1)
p_E2 <- get_prob(E2)
p_E3 <- get_prob(E3)

# Print individual probabilities clearly
cat(paste("Probability of E1:", p_E1,
          "\n", "Probability of E2:", p_E2,
          "\n", "Probability of E3:", p_E3))
```

```
## Probability of E1: 0.5
## Probability of E2: 0.5
## Probability of E3: 0.5
```

```

# Check Pairwise Independence (Intersections)
int_1_2 <- intersect(E1, E2)
int_1_3 <- intersect(E1, E3)
int_2_3 <- intersect(E2, E3)

cat(paste("Intersection of E1 and E2:", int_1_2,
          "\n", "Intersection of E1 and E3:", int_1_3,
          "\n", "Intersection of E2 and E3:", int_2_3))

## Intersection of E1 and E2: HH
## Intersection of E1 and E3: HH
## Intersection of E2 and E3: HH

# Check if P(A and B) == P(A) * P(B)
p_int_1_2 <- get_prob(int_1_2) == (p_E1 * p_E2)
p_int_1_3 <- get_prob(int_1_3) == (p_E1 * p_E3)
p_int_2_3 <- get_prob(int_2_3) == (p_E2 * p_E3)

cat(paste("Pair 1 & 2:", get_prob(int_1_2), "==", (p_E1 * p_E2),
          "\n", "Pair 1 & 3:", get_prob(int_1_3), "==", (p_E1 * p_E3),
          "\n", "Pair 2 & 3:", get_prob(int_2_3), "==", (p_E2 * p_E3)))

## Pair 1 & 2: 0.25 == 0.25
## Pair 1 & 3: 0.25 == 0.25
## Pair 2 & 3: 0.25 == 0.25

# Check Mutual Independence (Intersection of all three)
int_all <- intersect(intersect(E1, E2), E3)
p_int_all <- get_prob(int_all)
p_product_all <- p_E1 * p_E2 * p_E3
is_mutually_independent <- p_int_all == p_product_all

# Print results
cat(paste("E1 & E2 independent?:", p_int_1_2,
          "\n", "E1 & E3 independent?:", p_int_1_3,
          "\n", "E2 & E3 independent?:", p_int_2_3,
          "\n", "Mutually independent?:", is_mutually_independent))

## E1 & E2 independent?: TRUE
## E1 & E3 independent?: TRUE
## E2 & E3 independent?: TRUE
## Mutually independent?: FALSE

```

Conclusion

The simulation and mathematical analysis demonstrate that events E_1 , E_2 , and E_3 are **pairwise independent but not mutually independent**.

When tested two at a time, knowing the outcome of one event leaves the probability of the other event completely unchanged at 50%. This satisfies the pairwise multiplication rule where $P(A \cap B) = P(A) \times P(B) = 0.25$.

However, they fail the test for mutual independence because the events share a strict logical relationship. If we know that the first coin is Heads (E_2) and the second coin is Heads (E_3), the probability that the two coins match (E_1) instantly jumps from 50% to 100%. Because combining our knowledge of two events completely eliminates the randomness of the third event, the group multiplication rule breaks ($P(E_1 \cap E_2 \cap E_3) = 0.25$, which does not equal $P(E_1) \times P(E_2) \times P(E_3) = 0.125$).